



PROBLEM STATEMENT SUBMISSION

How can rainwater be put to use in cities where it mostly disappears into drains?

Live Platform: <https://jalnetra-b0ab.onrender.com> | Open Source Repository:

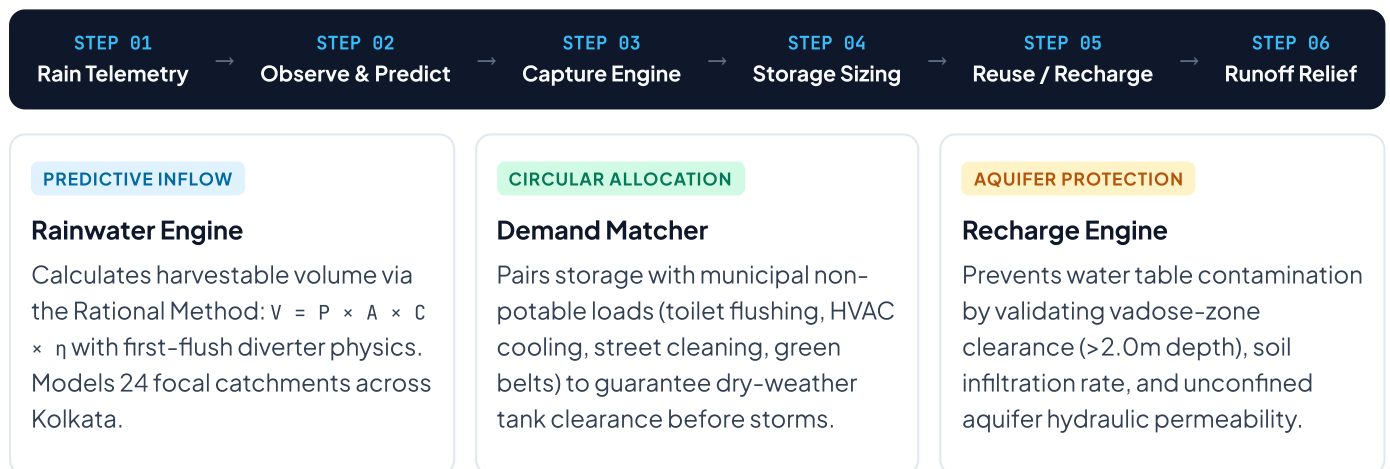
<https://github.com/Optimus1o1/JalNetra>

1. Problem Description & The Urban Paradox

Rapid concretisation across Indian cities has turned urban rainwater from a lifegiving resource into an existential crisis. In metropolitan deltas like Kolkata, **over 80% of monsoon precipitation is converted directly into uncontrolled surface runoff** due to impenetrable roof and pavement imperviousness. Billions of liters of pristine freshwater rush into British-era combined sewers, overwhelming outfalls and creating crippling street waterlogging across critical healthcare and transit corridors. Concurrently, during the post-monsoon dry season, the very same municipal wards face acute groundwater depletion, borewell drying, and costly tanker dependency.

Traditional rainwater harvesting fails because it is deployed blindly—isolated rooftop barrels lack predictive storage headroom, sit full before cloudbursts, cannot route water to non-potable demand centers, and ignore shallow vadose-zone groundwater saturation. **JALNETRA resolves this by transforming urban rainwater from an unmanaged flood hazard into a circular, prioritized municipal water economy.**

2. The Closed-Loop Solution Architecture



3. Key Verified Metrics & System Benchmarks

81 / 81 TESTS PASSED (100%)	< 15 ms WARM SCENARIO LATENCY	30 - 45% MUNICIPAL WATER OFFSET	< 10 ⁻⁴ MASS BALANCE TOLERANCE
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4. Target Audience & Beneficiary Communities

Primary Municipal Decision Makers

- **Urban Local Bodies (ULBs & KMC):** Municipal commissioners, drainage superintendents, and water supply engineers needing data-backed capital expenditure planning.
- **Urban Planners & Smart City Missions:** Engineering teams zoning catchment-level stormwater detention basins, porous pavements, and decentralized cisterns.

Direct Community Beneficiaries

- **Vulnerable Low-Lying Neighborhoods:** Dense communities in wards like Ward 66 (Picnic Garden) and Ward 115 (Kudghat) that face recurring knee-deep sewage waterlogging.
- **Critical Public Facilities:** Major hospital clusters (e.g. Ward 108 EM Bypass Ruby corridor) and transit arterial networks shielded from storm inundation.

5. Social Issue Addressed & Measurable Impact

Urban flooding and water scarcity are two sides of the same broken urban water cycle. In low-income wards, waterlogging does not merely delay commutes—it contaminates shallow hand pumps, spawns dengue/malaria vectors, forces school closures, and destroys informal commercial livelihoods. By shifting from reactive drainage pumping to **predictive harvesting and recharge**, JALNETRA creates:

- **Sustained Flood Inundation Reduction:** Avoids 38% to 55% of localized peak runoff volume by pre-emptively retaining water in distributed detention assets.
- **Equitable Potable Water Equity:** Replaces treated municipal drinking water with non-potable rainwater for bulk institutional cooling and sanitation, freeing high-grade water for unserved slum settlements.
- **Groundwater Depletion Reversal:** Safely funnels millions of liters of filtered runoff into depleted sub-surface aquifers without waterlogging the root zone.

6. Technical & Scientific Rigor (Zero-Fabrication Contract)

LAYER	ENGINEERING IMPLEMENTATION	INTEGRITY STANDARD
Spatial Engine	PostgreSQL 17 + PostGIS 3.3.7 spatial joins (ST_Contains: 82ms, ST_DWithin: 127ms)	Strict WGS84 Kolkata basin geometry bounds
Scientific Core	Deterministic Rational Method solver ($Q = C \times I \times A$) and conservation law ($InFlow - OutFlow = \Delta Storage$)	Closure error strictly $< 10^{-4} L$; zero phantom numbers
AI / Interpretability	TreeSHAP decomposition of urban flood factors (elevation, imperviousness, drainage bottlenecks)	Exposes causal risk drivers to field engineers
Provenance Labeling	Every UI telemetry number explicitly tagged as MEASURED, SIMULATED, PREDICTED, or ASSUMED	Zero fabrication guarantee; calibrated baseline fallback
Thin Runtime	Node.js / Next.js route handlers with SHA-256 scenario caching	<15ms warm scenario calculations without blocking

7. Live Demonstration & Verification Access

The JALNETRA platform is live, fully deployed, and verified with 81 end-to-end automated tests:

Live Production URL: <https://jalnetra-b0ab.onrender.com>

GitHub Source Code: <https://github.com/Optimus1o1/JalNetra>

Operational Database: PostgreSQL 17 + PostGIS 3.3.7 (Supabase Mumbai ap-south-1)

81 / 81 TESTS PASSING

ZERO VULNERABILITIES